# SVM Multiclass Prediction

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**Abstract.** This study evaluates SVM architecture across a tabular domain containing employee evidence. In this classification task, the SVM prioritized stability ($C = 1$) but remained at 71% accuracy due to a class imbalance.

**Keywords:** SVM, Multiclass Classification, Salary Insights, Career Stage Prediction, Supervised Learning

## 1 Introduction

This study presents a multiclass classification approach using Support Vector Machines (SVM) to predict employee experience levels. Leveraging a dataset of salary and job role attributes, the project explores the efficacy of kernel-based learning in demographic classification.

## 2 SVM Classification Model

The goal of this project is to predict an employee's experience level (*Entry*, *Mid*, *Senior*, *Executive*) using an SVM classifier. The model is trained on the "Salary Insights by Job Role" dataset (16534 samples and 11 features).

### 2.1 Dataset Description and Preparation

The raw dataset includes 11 features: **work_year**, **experience_level**, **employment_type**, **job_title**, **salary**, **salary_currency**, **salary_in_usd**, **employee_residence**, **remote_ratio**, **company_location**, and **company_size**.

**Data filtering** Data quality was improved by removing redundant columns (**salary** and **salary_currency**) and limiting the scope to full-time employees (over 99% work full-time). The **job_title** column contained a substantial number of rare categories, which introduced noise into the dataset. To mitigate this, positions with fewer than ten samples were grouped into a new category labeled "Other". Furthermore, **employee_residence** and **company_location** exhibit very similar patterns with respect to salary, making one of them redundant. Thus, **employee_residence** was retained and filtered to remove rare categories.

**Encoding and Partitioning** Features were categorized by their mathematical relationship to the target:

- **Continuous**: **salary_in_usd** was Z-standardized to act as the primary discriminative axis.
- **Categorical (OHE)**: **work_year** and **remote_ratio** (0, 50, 100) were one-hot encoded to capture discrete economic shifts and non-linear work-model influences rather than forced linear/ordinal trends. Remaining attributes (**job_title**, **employee_residence**, **company_size**) were also encoded via OHE.
- **Target**: **experience_level** was preserved as a 1D categorical vector for compatibility with the SVM's multiclass strategy.

The data was partitioned using a **stratified 80/20 split**[1] , ensuring that minority classes (EN, EX) remained proportionately represented.

### 2.2 Mathematical Description of the Model

The SVM classifier operates by finding an optimal separating hyperplane that maximizes the margin between the four distinct *experience_level* classes in the feature space.While a linear relationship exists between the target variable and the primary feature (*salary_in_usd*), the complete feature space—which includes the one-hot encoded *job_title* and the non-monotonic *remote_ratio*—introduces significant complexity. This requires a non-linear approach to achieve optimal separation.Specifically, the model uses the Radial Basis Function Kernel to employ the Kernel Trick. This implicitly maps the input data from its original **89**-dimensional space into a higher-dimensional feature space where the non-linear boundaries (such as the salary overlap between MI and SE roles) become linearly separable.For a training set $\{\mathbf{x}_p, y_p\}_{p=1}^{P}$, the model's goal is to minimize the objective function[2] , defined as:

$$\min_{\mathbf{w},b,\xi} \quad \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{p=1}^{P}\xi_p$$

subject to the constraints:

$$y_p(\mathbf{w}\cdot\phi(\mathbf{x}_p)+b) \geq 1-\xi_p \quad \text{and} \quad \xi_p \geq 0$$

Here, $\frac{1}{2}\|\mathbf{w}\|_2^2$ represents the term used to maximize the margin width $(\frac{2}{\|\mathbf{w}\|})$[3] ,while $\xi_p$ are *slack variables* that measure how much a specific point violates the margin[2] .The hyperparameter $C$ acts as a penalty, balancing the trade-off between a wide margin and misclassification errors.The feature mapping function $\phi(\mathbf{x}_p)$, defined by the RBF kernel, ensures the model captures the high-variance non-linearities required to differentiate overlapping experience levels[2] .

### 2.3 Training Procedure

- **Search Strategy**: **GridSearchCV**[1] was employed to evaluate combinations of the regularization parameter ($C$) and the kernel coefficient ($\gamma$). This allowed to navigate the bias-variance tradeoff by testing values of $C \in \{1, 5, 8, 10, 15\}$ and $\gamma \in \{\text{'scale'}, 0.001, 0.01, 1\}$.
- **Cross-Validation**: To ensure results were not specific to a single data split, a **5-fold stratified cross-validation** was utilized, maintaining class proportions.

### 2.4 Experimental Results

The SVM model underwent several iterations of hyperparameter tuning using **GridSearchCV**. The initial search iterations focused on a broader range of regularization strengths, which progressively narrowed toward a specific range of values.

- **Search History**: The grid search iterations consistently identified high-regularization values as the best performers, with the optimal **C** value evolving from **10**, to **8**, and finally to **5** in the most refined parameter grid: $C \in \{1, 5, 8, 10, 15\}$ and $\gamma =$ 'scale'.
- **Optimal Parameters**: The final execution of the search strategy identified the following configuration as the most effective: **'C': 5, 'gamma': 'scale', 'kernel': 'rbf'**, achieving a peak cross-validation accuracy of **70.75%**.

**Comparative Performance Analysis: $C = 5$ vs. $C = 1$** Following the grid search, a manual evaluation of $\mathbf{C = 1}$ was conducted to prioritize model generalization over training accuracy. While overall accuracy dropped slightly to 70.15% and the macro average F1-score decreased from 0.52 to 0.49, the weighted average was moderately reduced from 0.68 to 0.67. This stability is driven by the **SE** class recall, which rose to 0.94. However, this comes at the cost of minority class sensitivity, with recall for **EX** and **EN** falling to 0.22 and 0.32, respectively. The confusion matrix in Fig. 1 serves as a critical diagnostic metric, revealing significant overlap between seniority levels. Specifically, the frequent misclassification of **MI** as **SE** (517 observations) and **EX** as **SE** (69 observations) suggests that the the SVM defaults to the **SE** category due to the overlapping salary ranges and job titles across these experience levels.

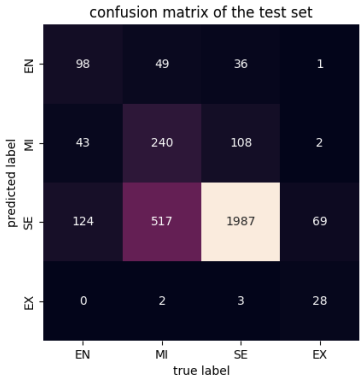


**Fig. 1.** Confusion Matrix ($C = 5$, $\gamma = \text{scale}$, kernel RBF)

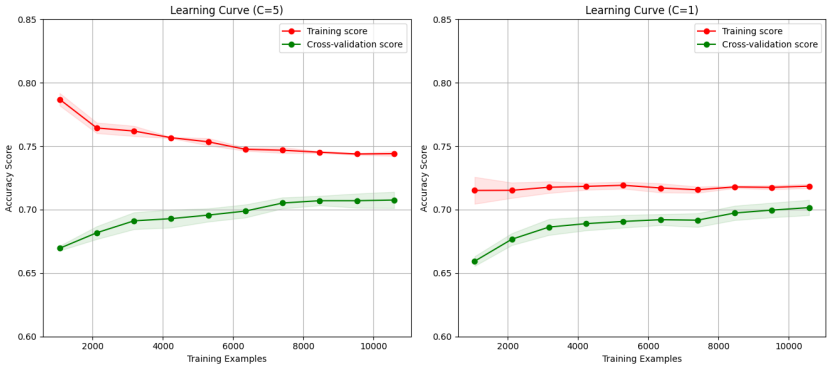


**Fig. 2.** Learning Curves ($C = 5$, $C = 1$)

### 2.5 Discussion

The transition from $\mathbf{C = 5}$ to $\mathbf{C = 1}$ highlights a critical trade-off: higher $C$ values maximize validation scores but create high variance (larger gap between learning curves), whereas a softer margin ($C = 1$) ensures model stability through learning curve convergence (see Fig. 2). Despite this tuning, accuracy remains constant at **71%** due to **inherent class imbalance**, where the dominant **SE** class acts as a majority class bias in the confusion matrix (Fig. 1). This limit indicates that the overlap in salary distributions between seniority levels cannot be resolved by regularization alone.

## References

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